Graph traversal and connectivity exercises

A07-B01

For the fixed graph, write every adjacency list and its 8×8 binary adjacency matrix. Count list entries and matrix ones. Direct every edge from the smaller letter to the larger letter. How many list entries remain? Then add two parallel A→B arcs and one B→B self-loop. How many entries does an ordinary list representation contain, and how many ones does the binary matrix contain?

A07-B02

Run BFS from A on the fixed undirected graph, scanning neighbors alphabetically. Give the initial Q and, after every completed scan, Q, newly discovered vertices, gray vertices, and black vertices. Give final d, π, and the parent path from A to G. Explain why C does not update D or E.

A07-B03

Reverse every adjacency list of the fixed graph. Give the dequeue order, every d and π value, and the parent path from A to G. Identify the parents that change and prove that neighbor order cannot change distances.

A07-B04

Prove that distances in FIFO order are nondecreasing and differ by at most 1. Then prove minimum-edge correctness, including the bound supplied by parent paths. Replace FIFO with LIFO while retaining first-discovery labels. Does correctness survive? Use undirected edges {sa,sb,ax,xy,yt,bt}, source s, alphabetical neighbor scans, pushes in that order, and pops from the back to give a counterexample.

A07-B05

Compare BFS from A and a complete search forest on the fixed graph. Count adjacency-list entries scanned, matrix cells probed, and enqueue operations. Give general time and extra-space bounds using n, m, and reachable count r. Initialize once and retain colors across trees. Why does a single isolated source show that matrix BFS is not always $\Theta(n^2)$ on every input?

A07-B06

In the directed graph V={s,a,b,c,t,z} with arcs {s→a,a→b,b→a,c→s,b→t}, run BFS from s with alphabetical neighbor order. Give d and π. If an outer white-root loop uses vertex order a,b,c,s,t,z, do its trees equal the strongly connected components? Finally compare minimum edge count and minimum total weight for arcs s→t of weight 10, s→x of weight 1, and x→t of weight 1.

A07-D01 Replay the execution

Use the fixed graph and orders to compute every d, f, and parent. List the stack after each discovery and finish. Explain whether scanning an edge advances the clock.

A07-D02 Classify every edge

List all nine edges in scan order, with target color and type. Why do A→C and D→C both reach BLACK but have different types? Can timestamps distinguish tree from forward?

A07-D03 White-path counterexample

Use u=C and v=D to refute “reachable implies descendant.” Classify interval pairs [2,9] with [5,8], and [3,4] with [10,11]. Prove [1,7] and [4,9] cannot coexist in a DFS.

A07-D04 Undirected model

An undirected graph contains only {p,q}, represented at both endpoints. What happens if every edge to GRAY reports a cycle? Repair the test for simple graphs and for multigraphs.

A07-D05 Prove both cycle directions

For finite directed graphs allowing self-loops, prove that a cycle exists iff full DFS has a back edge. Specify the white-path snapshot. Give two cycle certificates in the fixed graph.

A07-D06 DFS versus Kahn

Delete C→A and F→E. Compute reverse finish order and alphabetical-minimum Kahn order. Check every edge in both orders. Restore C→A: which vertices remain at Kahn termination? Explain with remaining indegrees.

Connectivity exercises

A07-C01

Find the SCC partition and condensation of G. Explain why one original DFS tree is not a component.

A07-C02

Run both Kosaraju passes alphabetically. Give all first-pass d/f values and discovery order in each second-pass tree.

A07-C03

Prove that an inter-SCC arc C→D gives F(C)>F(D), and use it to prove pass two cannot merge components.

A07-C04

Run Tarjan on a→b,b→a,a→c,c→b. Why cannot onstack be replaced by gray?

A07-C05

Give disc/low, cuts and bridges for the undirected example. Explain the root exception and equality case.

A07-C06

List every edge-stack pop batch. Explain why the boundary tree edge must be included.

A07-C07

Delete c, d, cd, and de separately. List the resulting components and counts.

A07-C08

Two parallel edges join u and v. Why does skipping a parent vertex misclassify a bridge? Fix it, and distinguish CLRS bcc from this section’ s block convention.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

CLRS, Introduction to Algorithms, 3rd ed., §§22.1–22.5 and Problem 22-2. Bridges are included here as one-edge blocks. Original handwritten notes 演算法_wjungle, PDF pp. 34–36, 40, for DFS notation and BFS augmenting paths. MIT 6.046J Fall 2004, L13 pp. 2–6 and L14 pp. 43–59; Fall 2005, lec16 pp. 2–5 and lec17 pp. 43–59. Representation and BFS background. MIT 6.006 Fall 2011 Lecture 14 pp. 2–6; Spring 2020 Lecture 10 pp. 2–4. DFS and topological order. Robert Tarjan, “Depth-First Search and Linear Graph Algorithms,” SIAM J. Comput. 1(2), 146–160 (1972), doi:10.1137/0201010. SCCs.

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